

Criterion	Rules	Conceptual Diagram	Polyomino Example
Criterion 1: Conway's Criterion	(1) Sides a and d will be translates of each other. (2) Sides b, c, e, f will be centrosymmetric.		
Criterion 2	(1) Sides a and d will be translates of each other. (2) Sides b and c will be centrosymmetric. (3) Sides e and f will glide reflect to each other.		
Criterion 3	(1) Sides a and d will be translates of each other. (2) Sides b and c will glide reflect to each other. (3) Sides e and f will glide reflect to each other.		
Criterion 4: Translation Criterion	(1) Sides a and d will be translates of each other. (2) Sides b and e will be translates of each other. (3) Sides c and f will be translates of each other.		
Criterion 5	(1) Sides a and d will be translates of each other. (2) Sides b and f will glide reflect to each other. (3) Sides c and e will glide reflect to each other. (4) The glide reflections above will be parallel.		
Criterion 6	(1) Sides a and d will glide reflect to each other. (2) Sides b and f will glide reflect to each other. (3) Sides c and e will be centrosymmetric. (4) The glide reflections above will be perpendicular.		
Criterion 7	(1) Side c will rotate 120° around D to side d . (2) Side e will rotate 120° around F to side f . (3) Side a will rotate 120° around B to side b .		Impossible.
Criterion 8	(1) Side c will rotate 120° around D to side d . (2) Side e will be centrosymmetric. (3) Side a will rotate 60° around B to side b .		Impossible.
Criterion 9	(1) Side b will rotate 90° around C to side c . (2) Side a will be centrosymmetric. (3) Side d will rotate 90° around E to side e .		

Table 58: The 9 ways a tile can tile the plane isohedrally. According to [Langerman and Winslow \(2015, Section 3\)](#) these are first shown in [Heesch and Kienzle \(2013, Table 10\)](#), and reproduced in [Schattschneider and Escher \(1990, p. 326\)](#). The final rules in Criteria 5 and 6 were added by [Church \(2008, p. 28\)](#), who discovered counterexamples that show the extra rules are needed.